EXPERIMENT 10

Web Performance and Security Audit using Lighthouse

AIM

To audit and improve a website's performance, accessibility, best practices, SEO and security using Google Lighthouse.

ALGORITHM

1. Open the website in Google Chrome.
2. Run Lighthouse from Chrome DevTools.
3. Record the scores for Performance, Accessibility, Best Practices and SEO.
4. Optimize images, scripts, semantic HTML, metadata and security settings.
5. Run Lighthouse again and compare the scores.

PROGRAM

```
<!-- Example performance and accessibility improvements -->
<img src="photo.webp" alt="Product image" loading="lazy">

<meta name="viewport" content="width=device-width, initial-scale=1">
<meta name="description" content="A responsive web application">

<!-- Load non-critical JavaScript without blocking HTML -->
<script src="app.js" defer></script>

/* Example: enable compression and security headers in Express */
const helmet=require('helmet');
const compression=require('compression');
app.use(helmet());
app.use(compression());

// Run an audit:
// Chrome -> DevTools -> Lighthouse -> Analyze page load
// Compare Performance, Accessibility, Best Practices and SEO scores.
```

RESULT

Thus, the website was audited using Lighthouse and common performance, accessibility, SEO and security improvements were identified and applied.

VIVA QUESTIONS

1. What is Google Lighthouse?
2. Name the main Lighthouse audit categories.
3. What is lazy loading?
4. Why are alt attributes important?
5. Why should a website use HTTPS and security headers?